

# SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-24 22:27 UTC

## SEC P vs NP — notebook summary

Generated 2026-04-24 22:27 UTC

- Cycles recorded: **46**
- Time span: 24.8h (~1.86 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp\_notebook.jsonl

### Verdict distribution

| Verdict      | Count |
|--------------|-------|
| INCONCLUSIVE | 35    |
| FALSIFIED    | 8     |
| SUPPORTED    | 3     |

### Mathematical fields explored (field\_A)

| Field                                                                 | Cycles |
|-----------------------------------------------------------------------|--------|
| Noncommutative geometry                                               | 2      |
| Combinatorial homotopy theory                                         | 1      |
| Kazhdan-Lusztig theory of Hecke algebras                              | 1      |
| Combinatorial algebraic topology                                      | 1      |
| Knot theory (quantum invariants)                                      | 1      |
| Quadratic forms over finite fields                                    | 1      |
| (Grothendieck-Witt groups)                                            |        |
| Clifford algebras over finite fields                                  | 1      |
| Toric geometry (via Gröbner degenerations)                            | 1      |
| Arithmetic geometry (Tate-Shafarevich groups)                         | 1      |
| Geometric Langlands correspondence                                    | 1      |
| Algebraic graph theory (chromatic polynomials)                        | 1      |
| Clifford algebras over real vector spaces                             | 1      |
| Algebraic cycles (Chow groups over finite fields)                     | 1      |
| Algebraic lattice theory (Möbius functions of<br>flow lattices)       | 1      |
| Directed algebraic topology (d-space homology)                        | 1      |
| Motivic integration (Denef-Loeser zeta<br>functions)                  | 1      |
| Matroid theory (Clifford index of binary<br>matroids)                 | 1      |
| Representation theory of finite groups<br>(Frobenius-Schur indicator) | 1      |

| Field                                                          | Cycles |
|----------------------------------------------------------------|--------|
| Categorification and knot Floer homology                       | 1      |
| Modular forms                                                  | 1      |
| Algebraic geometry (ideals in polynomial rings)                | 1      |
| Hypergraph Tutte polynomials                                   | 1      |
| Galois theory of finite field extensions                       | 1      |
| Combinatorial homotopy theory (unstable 2-cohomotopy)          | 1      |
| Geometric group theory (Kazhdan's property T)                  | 1      |
| Quadratic forms over $\text{GF}(2)$                            | 1      |
| Commutative algebra (Castelnuovo-Mumford regularity)           | 1      |
| Algebraic K-theory (Bass Nil-groups)                           | 1      |
| Arithmetic geometry (Selmer groups of elliptic curves)         | 1      |
| Toric geometry (via initial ideals and monomial degenerations) | 1      |

### Last 15 cycles (chronological)

| Time                 | Verdict      | Title                                                              |
|----------------------|--------------|--------------------------------------------------------------------|
| 2026-04-24 11:55 UTC | INCONCLUSIVE | K-theory and SAT                                                   |
| 2026-04-24 13:08 UTC | FALSIFIED    | Convex Hull Facet Count and Resolution Proof Size                  |
| 2026-04-24 13:38 UTC | INCONCLUSIVE | Quantum Dimension of Clause Algebra Bounds                         |
| 2026-04-24 16:16 UTC | INCONCLUSIVE | Communication Complexity                                           |
| 2026-04-24 16:30 UTC | INCONCLUSIVE | Minimal Ideal Generators and Resolution Proof Size                 |
| 2026-04-24 17:19 UTC | INCONCLUSIVE | Vandermonde Rank of Clause-Variable Incidence Matrix Predicts Res  |
| 2026-04-24 18:09 UTC | INCONCLUSIVE | Average Sensitivity of Clause-Indicator Function                   |
| 2026-04-24 18:43 UTC | INCONCLUSIVE | Bounds Resolution                                                  |
| 2026-04-24 19:18 UTC | INCONCLUSIVE | Minimal ABP Size and Permutation Polynomial Degree                 |
| 2026-04-24 19:54 UTC | INCONCLUSIVE | Hypergraph Discrepancy and Communication Complexity of 3-SAT       |
| 2026-04-24 20:34 UTC | INCONCLUSIVE | Hilbert Function of Ideal Bounds Communication Complexity          |
| 2026-04-24 21:05 UTC | INCONCLUSIVE | Matrix Rigidity Bounds Communication Complexity of 3-SAT Instance  |
|                      |              | Ideal Generator Count and Bounded Arithmetic Proof Size            |
|                      |              | Orbit Signature Decay Implies Communication Entropy Barrier Growth |

| Time                 | Verdict      | Title                                                                        |
|----------------------|--------------|------------------------------------------------------------------------------|
| 2026-04-24 21:22 UTC | INCONCLUSIVE | ACC <sup>0</sup> Circuit Size Bounded by Finite Field Equation Solution Coun |
| 2026-04-24 22:09 UTC | FALSIFIED    | Average-Case SAT Hardness Linked to Finite Field Solution Counts             |
| 2026-04-24 22:27 UTC | INCONCLUSIVE | Specht Module Dimensions Bound ABP Size for Permutation Polynomia            |

## How to read the reports

- reports/supported\_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified\_findings.md — FALSIFIED with counterexamples.
- reports/daily\_YYYY-MM-DD.md — chronological log by day.
- pvsnp\_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

## PDF export

Install pandoc + a LaTeX engine once (sudo apt install pandoc texlive-xetex), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```